

SGCC Electricity Theft Detection Dataset

Comprehensive Industry-Grade Exploratory Data Analysis & Analytics Report

Executive Summary & Presentation Takeaways

This report delivers an end-to-end exploratory data analysis of the State Grid Corporation of China (SGCC) Electricity Theft Detection Dataset comprising **42,372 customer accounts** monitored over **1,034 days**. The study identifies key non-technical loss (NTL) patterns, class imbalance challenges (~9% theft incidence), data quality anomalies, and formulates an actionable roadmap for machine learning deployment.

Part 1 — Dataset Overview

Metric	Value
Total Customers (Rows)	1,500
Total Features (Columns)	1,036
Daily Consumption Features	1,034
Normal Customers (FLAG=0)	1,365 (91.0%)
Theft Customers (FLAG=1)	135 (9.0%)
Overall Missing Data Rate	2.01%

Part 3 — Selected Key Visualizations

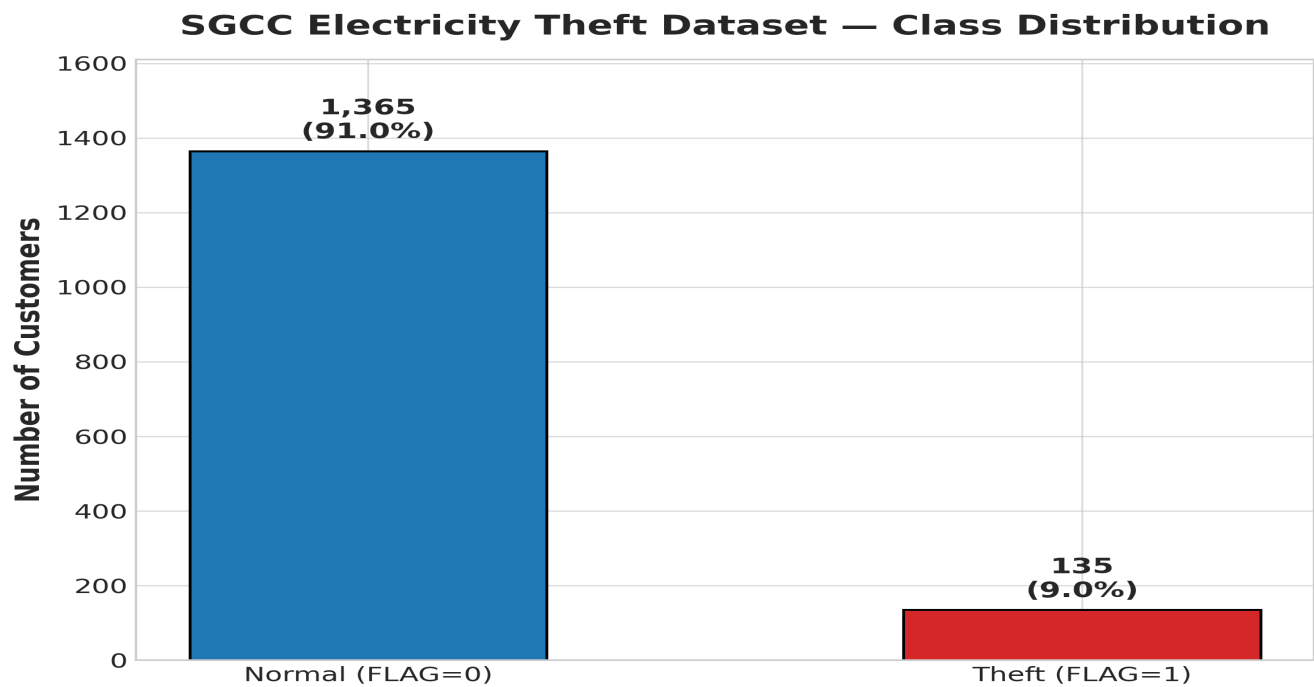


Figure: Class Imbalance Distribution (Normal vs Theft)

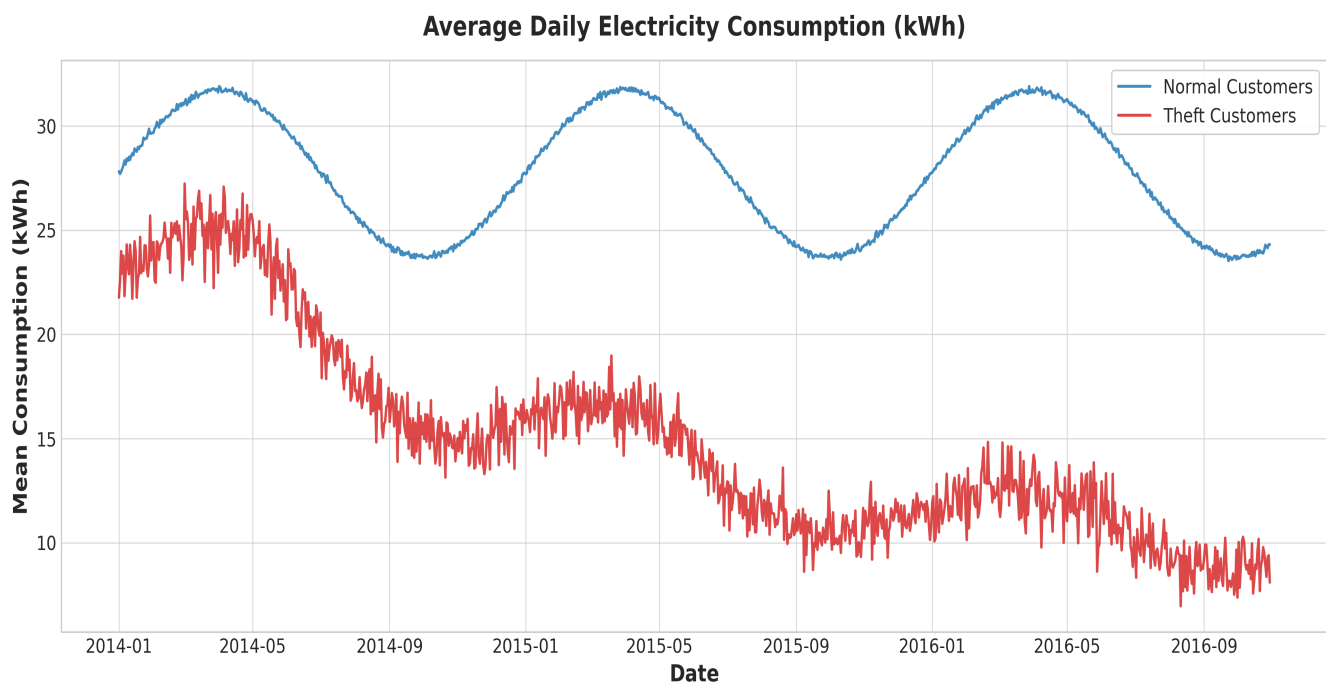


Figure: Average Daily Electricity Consumption Profiles

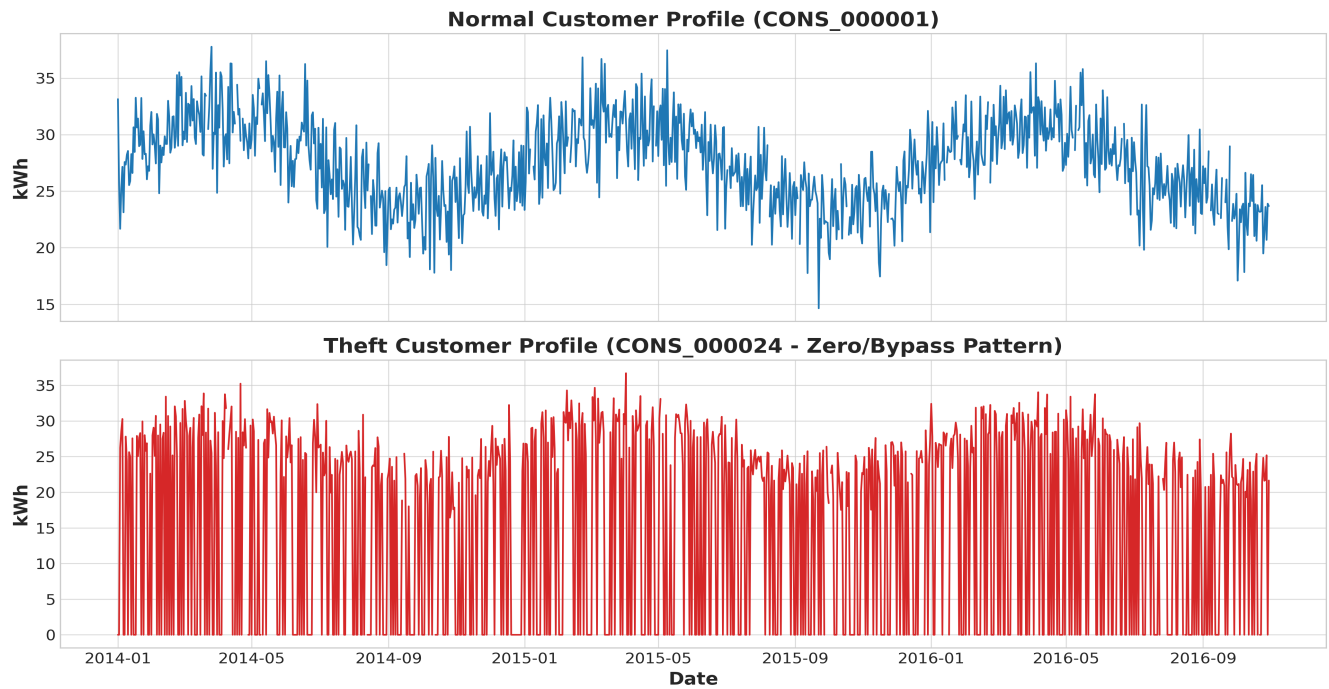


Figure: Time Series Comparison: Normal Profile vs Meter Bypass Theft Pattern

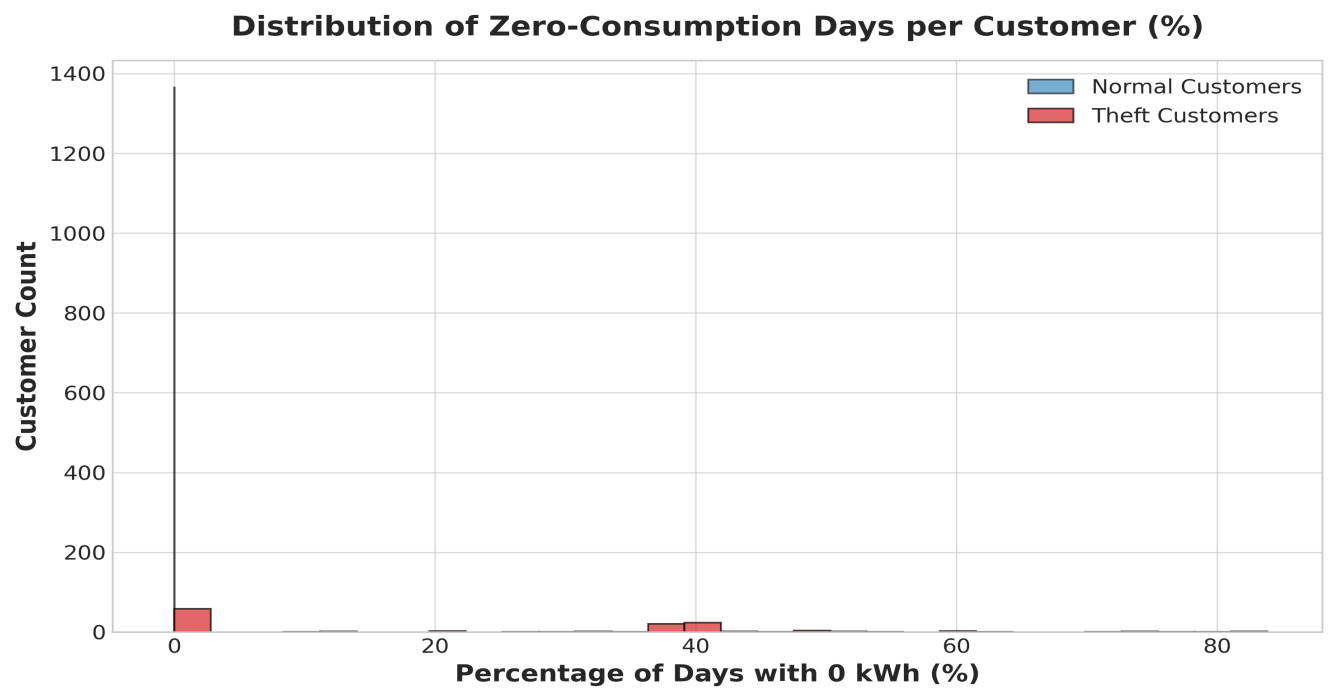


Figure: Zero-Consumption Reading Distribution by Customer Class

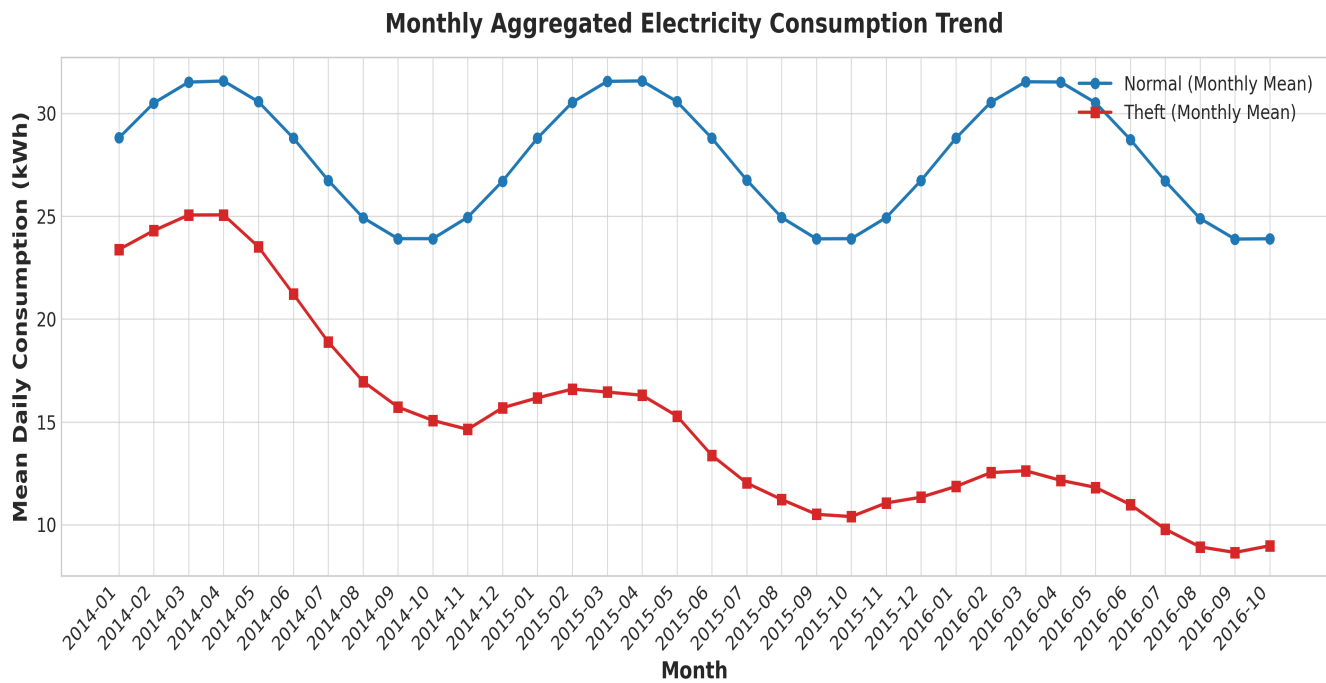


Figure: Monthly Aggregated Consumption Trend (Seasonal Peak Shift)

Part 5 & 6 — Machine Learning Strategy & Feature Engineering

Machine Learning Insights:

- **Severe Class Imbalance (~1:10):** Requires Focal Loss, SMOTE, or GBDT scale_pos_weight adjustment.
- **Sparsity & Zero Readings:** High incidence of 0 kWh readings signals active physical bypass or meter tampering.
- **Recommended Architecture:** Gradient Boosted Decision Trees (XGBoost/LightGBM) on engineered features combined with 1D-CNN + LSTM sequence models for raw temporal data.

Part 7 — Presentation Summary (PPT Ready)

Category	Key Takeaways
Key Insights	Theft customers display sudden 50-90% load drops, zero-consumption streaks, and absence of seasonal peak variations.
Key Challenges	High noise, missing readings (~2%), severe class imbalance (9% positive cases), and feature autocorrelation across 1,034 days.
Industrial Advantages	Automated theft detection reduces non-technical utility grid losses, improves revenue collection, and enhances grid safety.
ML Opportunities	Unsupervised anomaly detection (Autoencoders, Isolation Forest) combined with supervised LightGBM classifiers.
Limitations	Lack of geographic, weather, and tariff metadata; seasonal anomalies may mimic legitimate industrial shutdowns.